

Week 1 • Day 7

Data Cleaning Report & Week 1 Review

Generated the EDA summary dashboard, wrote the baseline report, and reviewed all Week 1 work

DATE	INTERN NAME	PROJECT	MENTOR
Day 7 of 28 Week 1	_____	Customer Churn Prediction & LTV Engine	_____

■ Technologies Used Today

Python

Pandas

Matplotlib

Seaborn

PostgreSQL

Git

■ Today's Key Metrics

10

Charts Created Week 1

7

GitHub Commits

7,043

Clean Rows Ready

5

Key Predictors Found

■ Day Objectives

- Generate multi-panel EDA summary dashboard figure
- Write baseline_report.md with all Week 1 insights
- Verify all 7 daily commits exist on GitHub
- Confirm cleaned dataset is ready for Week 2 feature engineering
- Review and understand all concepts from Days 1–7

■ Tasks Completed Today

1. Created reports/generate_report.py — multi-panel summary dashboard script

2. Generated 6-panel figure: churn distribution, tenure histogram, monthly charges, contract churn, correlation heatmap, key metrics table
3. Saved reports/figures/00_eda_summary_dashboard.png at 150 DPI
4. Wrote reports/baseline_report.md — 3-page summary of all EDA findings
5. Verified all 7 daily commits on GitHub using 'git log --oneline'
6. Reviewed df.shape, column names, and data types of customers_clean table
7. Documented top 5 churn predictors from EDA: Contract, Tenure, InternetService, PaymentMethod, OnlineSecurity
8. Planned Week 2 feature engineering targets based on EDA findings
9. Updated README.md with Week 1 completion status
10. Committed all files and verified GitHub repository is current

■ Concepts Learned

- matplotlib.gridspec.GridSpec — how to create complex multi-panel figures
- Summary dashboards: organising multiple charts into a single coherent story
- How EDA findings directly translate into feature engineering decisions
- Git log: reviewing commit history to verify daily progress
- The importance of documenting insights WHILE exploring — not after
- Churn prediction is a binary classification problem with class imbalance

■ Tools / Software Used

Tool / Library	Version / Notes	Status
Matplotlib GridSpec	Built-in	✓ Used for multi-panel layout
Git	Existing	✓ 7 commits verified

■ Outputs / Deliverables Produced

- ◆ reports/figures/00_eda_summary_dashboard.png — 6-panel summary figure
- ◆ reports/baseline_report.md — complete Week 1 findings document
- ◆ Updated README.md with Week 1 status
- ◆ Total charts created this week: 10 (charts 00-09)
- ◆ Confirmed: 7 GitHub commits, one per day
- ◆ Week 1 complete — ready for ML in Week 2

■ Problems Encountered & Solutions

Problem Faced	How I Solved It
Multi-panel figure was too crowded — text overlapping	Added hspace=0.45 and wspace=0.35 to GridSpec to increase spacing between panels

■ GitHub Commit Record

Commit Message	Week1-Day7: EDA summary dashboard, baseline report, Week 1 complete — 7 commits verified
Branch	main
Files Changed	reports/generate_report.py, reports/figures/00_eda_summary_dashboard.png, reports/baseline_report.md
Commit Hash	(fill after push)

■ End-of-Day Submission Checklist

■ Summary dashboard generated	■ baseline_report.md written
■ 7 commits verified on GitHub	■ 10 total charts in reports/
■ customers_clean verified	■ Top 5 predictors identified
■ Week 2 plan documented	■ README.md updated
■ No missing values confirmed	■ Week 1 Day 7 commit done

■ Self Assessment

9/10 Self Rating	5 Hours Spent	Week 1, Day 7 Progress	7/28 Day in Program
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■ Key Learnings & Reflections

- ★ Week 1 complete: database set up, data cleaned, 10 charts created, 7 commits made
- ★ Top churn predictors identified from EDA: contract type, tenure, internet service, payment method
- ★ The dataset has class imbalance (26.5% churn) — must address in Week 2 modelling
- ★ Clean folder structure + daily commits = professional project from day one

■ Plan for Tomorrow

- Week 2 Day 1: Create 15+ new features from existing columns
- Build ratio features: MonthlyCharges/tenure
- Create service count features: total add-ons subscribed
- Create loyalty segment labels and contract risk scores

INTERN SIGNATURE	MENTOR SIGNATURE	DATE SUBMITTED